Project Design Phase
Solution Architecture

Date	04-10-2026
Team ID	23KD1A4233
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Solution Architecture

The ServiceNow-Powered Commuter Transit Platform is engineered on a modular, three-tiered enterprise cloud model designed to handle end-to-end ticketing operations seamlessly. Passengers initiate travel orders through an interactive Service Portal, which captures route parameters, schedule timestamps, and ticket quantities. The system's underlying application controllers evaluate route matrices asynchronously, execute server-side price validations, and dispatch automated pass fulfillment tasks without manual platform intervention.

Architectural Layers

- **User Interaction Layer (Presentation Tier):** Features a responsive Service Portal & Service Catalog UI designed for commuters to specify origin and destination nodes, select travel dates, and configure passenger counts with real-time field validation.

- **Business Logic & Workflow Layer (Application Tier):** Utilizes Catalog Client Scripts, GlideAjax APIs, and Script Includes for client-server asynchronous execution, calculating total dynamic fares instantly, enforcing business rules, and driving Flow Designer automation for task provisioning.

- **Data & Platform Services Layer (Persistence & Operations Tier):** Relies on custom ServiceNow Tables to maintain operational datastores, integrated with the Notification Engine for dispatch alerts, external QR Generator APIs for pass encoding, and Performance Analytics for executive reporting.

Solution Architecture Diagram:

